Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

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1 Angles, Radians, and the Unit Circle

A turning wheel connects angle, distance, speed, and position. This chapter develops those connections from a coordinate sketch to a model of a rider's height over time. Work through each example with pencil and paper. Pause at each check before reading its explanation at the end of the chapter.

Learning objectives. Describe signed rotations; convert between degrees and radians; distinguish arc length, displacement, angular velocity, and tangential speed; derive exact unit-circle coordinates; and connect circular motion with periodic graphs and a height model.

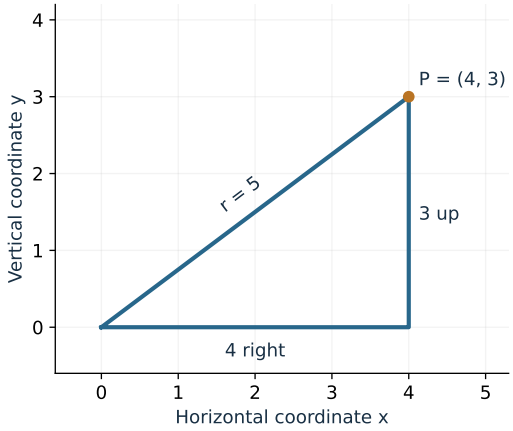
Online reading and interactive activities

Further reading: OpenStax, *Precalculus 2e*, 5.1 Angles and 5.2 Unit Circle: Sine and Cosine Functions.

1.1 Read a point before describing a turn

Imagine a marker on the rim of a turning wheel. We want to describe where the marker is, how far it has traveled, and how quickly it moves. These are different questions. Trigonometry connects them by treating an angle as a record of turning.

Begin with a coordinate plane. The origin is the point where the axes meet. In an ordered pair (x, y), the first number tells us horizontal position and the second tells us vertical position. Positive x means right; negative x means left. Positive y means above the horizontal axis; negative y means below it.



The coordinates describe two perpendicular moves. The distance from the origin is the hypotenuse.

A point and its distance from the origin

For $P = (4, 3)$, the Pythagorean theorem gives the distance r from the origin:

$$r^2 = 4^2 + 3^2 = 25, \qquad r = 5.$$

If we shrink this triangle until its hypotenuse is 1, every length is divided by 5. The new point is $(4/5, 3/5)$. It points in the same direction as $(4, 3)$, but lies on a circle of radius 1. This process is called *normalizing* the coordinates.

We will use fractions and square roots to describe some positions exactly. An expression such as $\sqrt{2}/2$ is an exact number. Its decimal approximation, about 0.707, is useful for estimating, but the rounded decimal loses some information. Both forms will appear in this module.

As you read, keep a sketch beside your calculations. A positive vertical coordinate should put the point above the horizontal axis; a distance should never be negative. These simple checks can catch many errors.

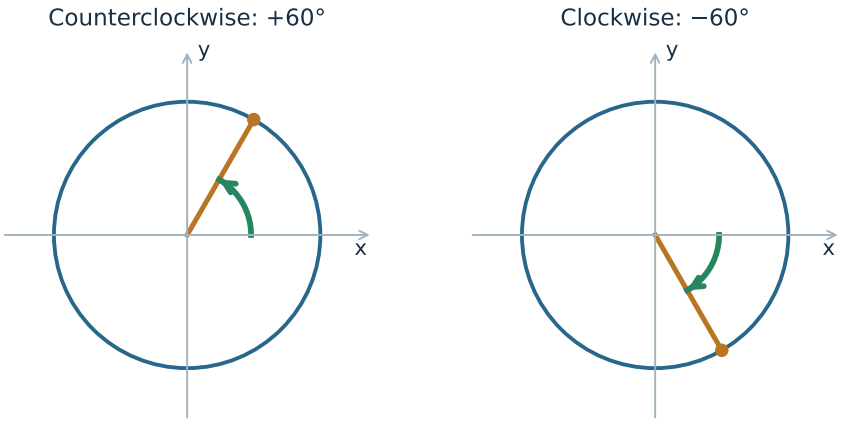
Check 1

A point is at $(-3, 4)$. Find its distance from the origin and divide both coordinates by that distance. Which signs should remain?

1.2 An angle records a signed rotation

An angle describes how a ray turns around its endpoint. A ray is a line that starts at one point and extends in one direction. In *standard position*, the endpoint is the origin and the starting ray points right along the positive x-axis. The final ray is called the *terminal ray*.

We use a sign to record the direction of turning. Counterclockwise turns are positive; clockwise turns are negative. A complete turn is 360°, a half-turn is 180°, and a quarter-turn is 90°. The Greek letter θ, pronounced "theta," often stands for an angle.



Begin at the positive horizontal axis. The sign tells you which way to turn.

Read the whole rotation

For 420°, turn counterclockwise through one full revolution and then another 60°:

$$420^\circ = 360^\circ + 60^\circ.$$

For -120°, turn clockwise through 120°. The terminal ray points down and left. You could reach that same ray by turning counterclockwise through 240°, but the two descriptions record different journeys.

A diagram usually shows the final direction more easily than it shows several complete turns. Do not

assume that an angle must be between zero and one revolution. A wheel may turn many times, and a negative angle is a perfectly valid description of motion.

Check 2

Where do $-90°$ and $450°$ end? Describe the direction and amount of turning for each.

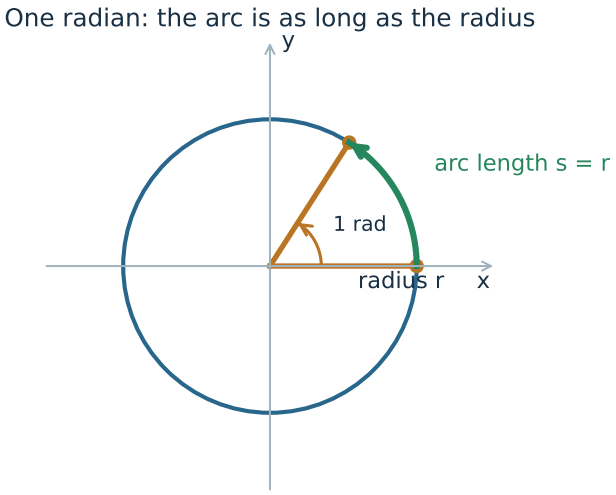
1.3 Radians compare an arc with a radius

Degrees divide a full turn into 360 equal pieces. Radians measure the same turning in a way tied directly to the circle. An *arc* is a curved part of a circle. For a central angle, the two rays meet at the circle's center and cut off an arc on its circumference.

Take the length of that arc and divide by the radius, using the same length units for both. The ratio is the angle's magnitude in radians:

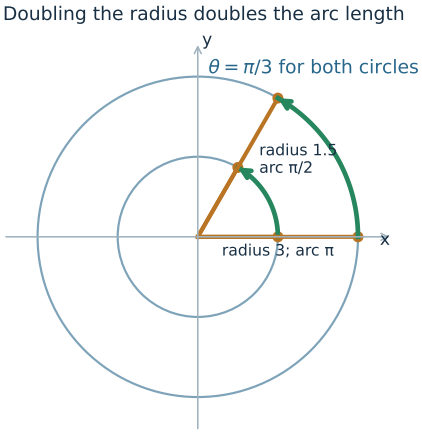
$$\boxed{\theta = \frac{s}{r}} \quad \text{(radian measure)}.$$

An angle of 1 radian cuts off an arc whose length equals the radius. It is a little less than $60°$: about $57.3°$. A radian is an angle unit; it is not another name for π.



One radian means one radius-length of arc.

Why does the size of the circle not change the angle? If you double the radius while keeping the same two rays, the arc also doubles. The ratio s/r stays the same. The picture below shows this for an angle of $\pi/3$ radians.



Angle describes turning. Radius describes size. The ratio of arc length to radius separates these ideas.

Why one turn is 2π radians

A circle's circumference is $2\pi r$. For a complete turn, divide the full circumference by the radius:

$$\theta = \frac{2\pi r}{r} = 2\pi.$$

Thus $360^\circ = 2\pi$ radians and $180^\circ = \pi$ radians. The number π is about 3.14159. A half-turn has radian measure π because its arc has length πr.

For clockwise motion, use a negative angle. If s means ordinary arc length, it is nonnegative and s/r gives the angle's magnitude. A signed angle requires an orientation as well. We will keep signed arc length and distance traveled separate.

Check 3

An arc is 9 cm long on a circle of radius 6 cm. What is the angle's magnitude in radians? At the same angle on a circle of radius 12 cm, how long is the arc?

1.4 Convert units without changing the angle

The relation $180^\circ = \pi$ radians lets us write the same angle in either unit. To convert, multiply by a fraction that equals 1 and cancels the unwanted unit.

$$\text{Degrees to radians: multiply by } \frac{\pi \text{ rad}}{180^\circ}.$$

$$\text{Radians to degrees: multiply by } \frac{180^\circ}{\pi \text{ rad}}.$$

Degrees to radians: 240°

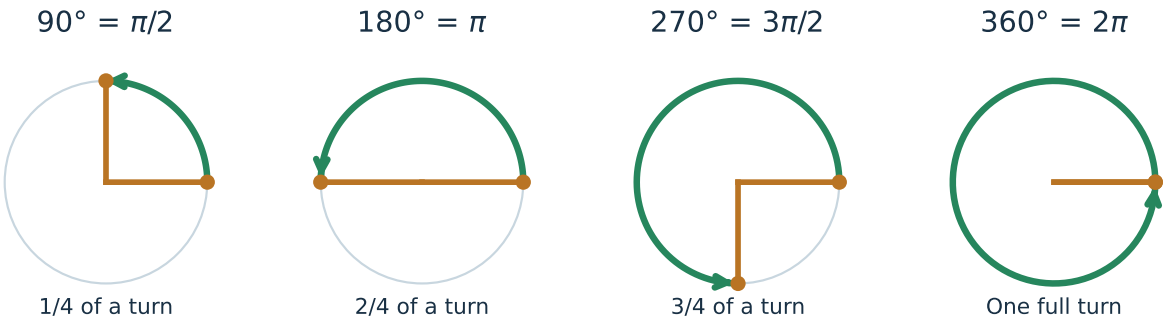
$$240^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = \frac{240\pi}{180} \text{ rad} = \frac{4\pi}{3} \text{ rad}.$$

The degree units cancel. Before calculating, we know that 240° is between a half-turn and a full turn. Our result, $4\pi/3$, is between π and 2π, as expected.

Radians to degrees: -5π/3

$$\left(-\frac{5\pi}{3} \text{ rad}\right) \frac{180^\circ}{\pi \text{ rad}} = -\frac{5(180^\circ)}{3} = -300^\circ.$$

The π factors and radian units cancel. Keep the negative sign: changing units does not change the direction of rotation.



A fraction of a full turn is the same fraction of 360° or of 2π radians.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	1/12
45°	$\pi/4$	1/8
60°	$\pi/3$	1/6
90°	$\pi/2$	1/4
180°	π	1/2
270°	$3\pi/2$	3/4
360°	2π	1

Leave an answer such as $4\pi/3$ exact unless a decimal is requested. To estimate it, use the calculator's π key and round at the end. Replacing π by 3.14 too early can add avoidable error.

Check the calculator's angle mode

In degree mode, $\sin(30) = 1/2$. In radian mode, the same turn is entered as $\sin(\pi/6) = 1/2$. Typing 30 in radian mode means 30 radians, a very different angle. Converting an angle and changing the calculator mode are separate actions.

Check 4

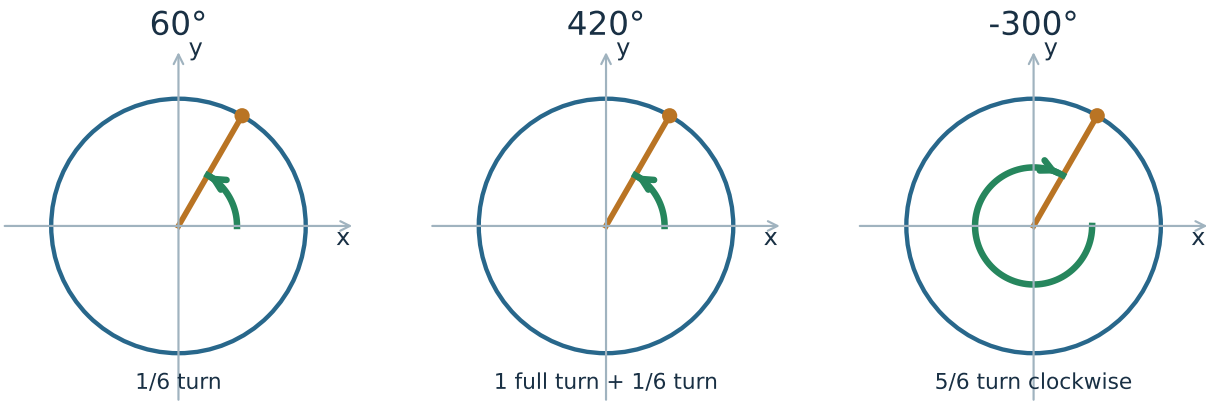
Convert $72°$ to radians and $7\pi/5$ radians to degrees. Check each result against one full turn.

1.5 Same final direction, different journeys

Two angles are *coterminal* when they have the same starting and terminal rays. Adding or subtracting a full turn changes the amount of rotation without changing the final direction.

$$\theta + 360°k \quad \text{or} \quad \theta + 2\pi k, \qquad k \in \mathbb{Z}.$$

The symbol $\mathbb{Z}$ means the integers: $\ldots, -2, -1, 0, 1, 2, \ldots$. Choose the formula that matches your angle unit. For example, $60°$, $420°$, and $-300°$ all end on the same ray.



A terminal ray does not tell us how many turns happened along the way.

Find a representative between 0 and 2π

Start with $-13\pi/4$. A full turn is $2\pi = 8\pi/4$, so add full turns using a common denominator:

$$-\frac{13\pi}{4} + \frac{8\pi}{4} = -\frac{5\pi}{4}, \qquad -\frac{5\pi}{4} + \frac{8\pi}{4} = \frac{3\pi}{4}.$$

Now $0 \le 3\pi/4 < 2\pi$. The terminal ray is in the upper-left part of the plane. We added two full turns; the original clockwise rotation was not erased from its physical history.

The interval $[0, 2\pi)$ includes 0 and excludes 2π. Those two angles point in the same direction, so

including both would repeat a representative. Similarly, use $[0°, 360°)$ for a single representative in degrees.

For the final point on a circle, a coterminal angle is enough. For distance traveled, use the actual rotation. A marker turning through 420° travels seven times as far as one turning through 60° on the same circle.

Check 5

Find the angle in $[0°, 360°)$ coterminal with 870°. How many full turns did you set aside?

1.6 Arc length, distance, and displacement

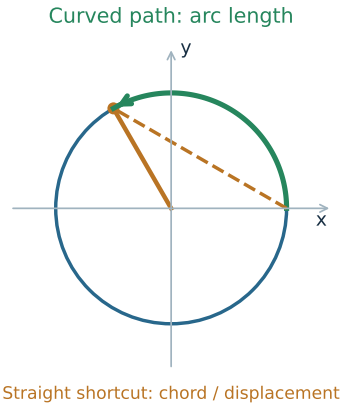
The equation $\theta = s/r$ can be rearranged to find arc length. When θ is a signed angle in radians, $r\theta$ gives a *directed* arc length:

$$s_{\text{directed}} = r\theta.$$

If the motion proceeds in one direction without reversing, the distance traveled is

$$\boxed{\text{distance} = r|\theta|}.$$

The absolute-value bars remove the sign. A clockwise journey still covers a positive distance. If motion reverses, calculate the distance for each part and add those distances. The final net angle alone will not tell you the whole distance.



Distance along the rim follows the arc. Displacement compares the starting and ending positions directly.

A clockwise turn on a radius-6 circle

A point turns clockwise through $120°$ on a circle of radius 6 cm. Convert first:

$$\theta = -120° \frac{\pi}{180°} = -\frac{2\pi}{3} \text{ rad.}$$

Then the directed arc is $6(-2\pi/3) = -4\pi$ cm, while the distance traveled is 4π cm, about 12.57 cm. The negative directed arc records clockwise orientation; it is not a negative distance.

Why radians are required

For a quarter-turn, $\theta = \pi/2$, so the arc is $r\pi/2$, one quarter of $2\pi r$. Putting 90 directly into $r\theta$ would give $90r$, which is much larger than the whole circumference. Convert degrees before multiplying.

Displacement is the change in position from the start to the finish. Its magnitude is the straight-line distance between the endpoints, not the arc length. After one complete revolution, a point has traveled a whole circumference but has zero displacement.

Check 6

A point on a circle of radius 2 m moves clockwise through half a turn. Find the signed angle, distance traveled, and magnitude of displacement.

1.7 Angular velocity and tangential speed

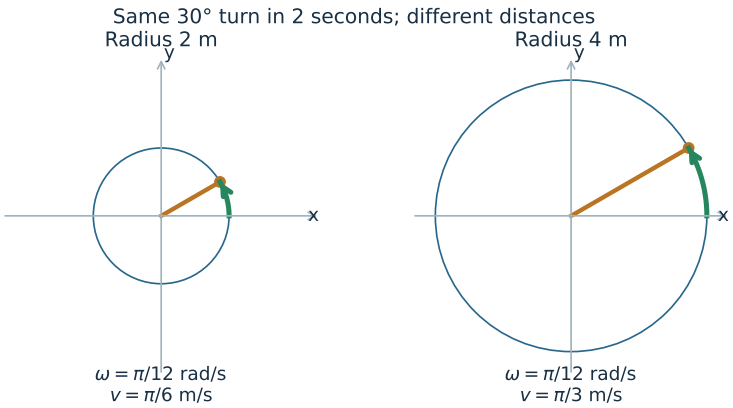
Angular velocity describes the rate at which the angle changes. Over a time interval of length Δt, the average angular velocity is

$$\omega = \frac{\Delta\theta}{\Delta t}.$$

The Greek letter ω is "omega," and Δ means "change in." With angles in radians and time in seconds, the units are radians per second, written rad/s. Counterclockwise angular velocity is positive; clockwise angular velocity is negative. Angular *speed* is the nonnegative magnitude $|\omega|$.

For uniform rotation, the rate is constant. A point farther from the center travels a longer arc during the same turn, so its tangential speed is greater:

$$\boxed{v = r|\omega|}.$$



The angular speed is shared by the whole wheel. Tangential speed depends on distance from the axis.

Convert revolutions per minute

A disk makes 6 counterclockwise revolutions per minute. A marker sits 0.5 m from the axis. Convert the angular rate one unit at a time:

$$6\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \frac{\pi}{5} \text{ rad/s}.$$

The revolutions and minutes cancel. The marker's tangential speed is $v = 0.5(\pi/5) = \pi/10$ m/s, about 0.314 m/s. A marker twice as far from the axis would move twice as fast.

The *period* T is the time needed for one complete revolution. For uniform rotation, angular speed is $2\pi/T$. If the rotation is counterclockwise, $\omega = 2\pi/T$; if clockwise, $\omega = -2\pi/T$. A smaller period means a faster rotation.

Average speed and instantaneous speed agree here because the rotation rate is constant. If the wheel speeds up and slows down, one average rate describes only the chosen interval.

Check 7

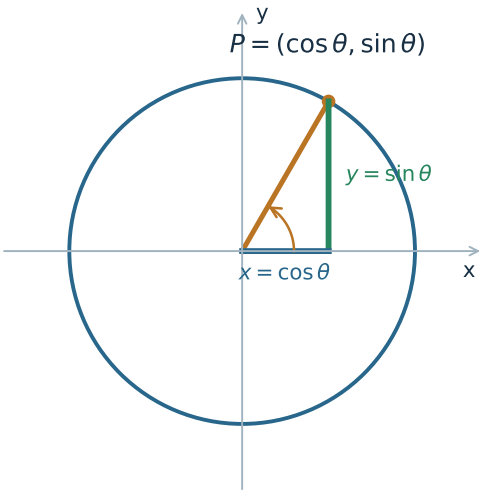
A wheel of radius 2 m makes one counterclockwise revolution every 20 seconds. Find its angular speed and tangential speed.

1.8 Sine and cosine are coordinates

The *unit circle* has center $(0, 0)$ and radius 1. Start at $(1, 0)$ and turn through angle θ. The coordinates of the resulting point define two functions:

$$\boxed{P = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate. Sine is the vertical coordinate. The expression $\sin\theta$ means "the sine of the angle θ," not "sin multiplied by theta." A function takes an input and assigns an output; here the input is an angle and the output is a coordinate.



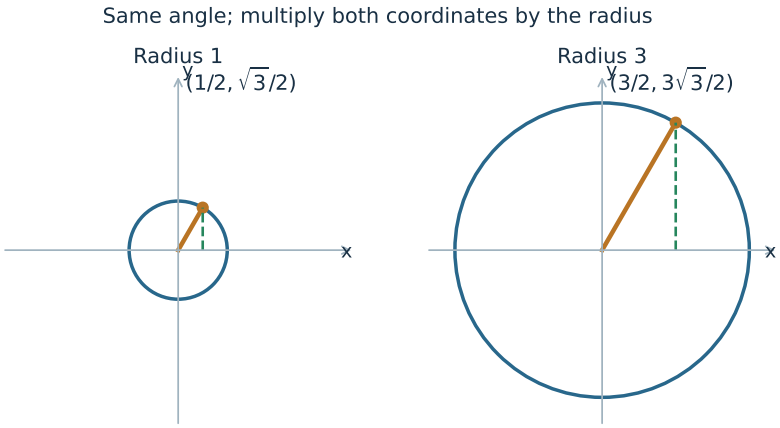
Keep the coordinate order: cosine first, sine second.

The familiar right-triangle ratios are built into this definition. In the first quadrant, the triangle has hypotenuse 1, adjacent side x, and opposite side y. Thus $\cos\theta = x/1 = x$ and $\sin\theta = y/1 = y$. The circle definition also handles angles beyond a right triangle, including negative angles and multiple turns.

Since every unit-circle point satisfies $x^2 + y^2 = 1$, its coordinates satisfy

$$\boxed{\cos^2\theta + \sin^2\theta = 1}.$$

The notation $\cos^2\theta$ means $(\cos\theta)^2$. Both coordinates lie between -1 and 1. These are useful checks on calculated values.



Changing the radius changes the physical coordinates, while sine and cosine of the angle stay the same.

Scale from a unit circle to a real wheel

If a point at angle θ has unit-circle coordinates $(3/5, 4/5)$, a point at that angle on a radius-5 circle has coordinates $(3, 4)$. In general, on a circle centered at the origin:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

These are physical coordinates. Dividing them by the radius recovers the unit-circle coordinates. The radius does not change the angle's sine or cosine.

Check 8

The unit-circle point at an angle is $(-3/5, -4/5)$. What is the corresponding point on a circle of radius 10 centered at the origin?

1.9 Derive exact values from two triangles

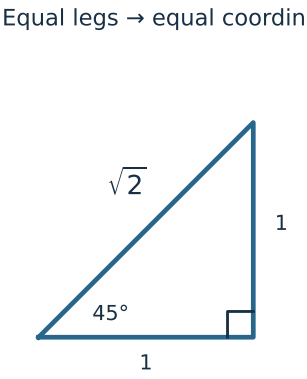
You do not need a separate rule for every special angle. Two right triangles supply the coordinate magnitudes for 30°, 45°, and 60°. The Pythagorean theorem explains where the square roots come from.

The 45° triangle

A right triangle with equal legs has two equal acute angles, each 45°. If each leg is 1, its hypotenuse is $\sqrt{1^2 + 1^2} = \sqrt{2}$. Divide every side by $\sqrt{2}$ to make the hypotenuse 1:

$$\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

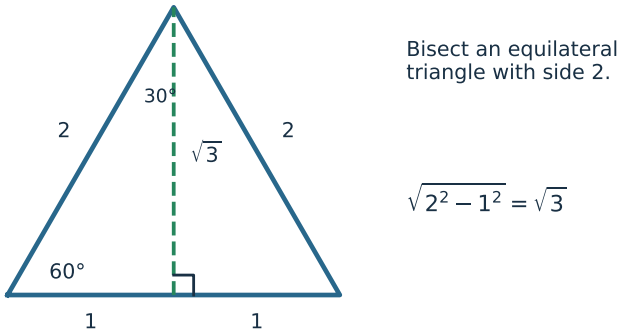
The last equality comes from multiplying top and bottom by $\sqrt{2}$. It changes the form of the number, not its value.



Equal legs give equal horizontal and vertical coordinates after normalization.

The 30° and 60° triangles

Cut an equilateral triangle of side 2 into two equal right triangles. Each has hypotenuse 2, short leg 1, and long leg $\sqrt{2^2 - 1^2} = \sqrt{3}$. The angles are 30°, 60°, and 90°.



Halving an equilateral triangle produces the ratio $1 : \sqrt{3} : 2$.

At 30°, the horizontal leg is $\sqrt{3}$ and the vertical leg is 1. Divide by the hypotenuse 2 to get $(\sqrt{3}/2, 1/2)$. At 60°, the roles of the two legs switch.

Angle	Radians	Cosine: horizontal	Sine: vertical
0°	0	1	0
30°	$\pi/6$	$\sqrt{3}/2$	$1/2$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
60°	$\pi/3$	$1/2$	$\sqrt{3}/2$
90°	$\pi/2$	0	1

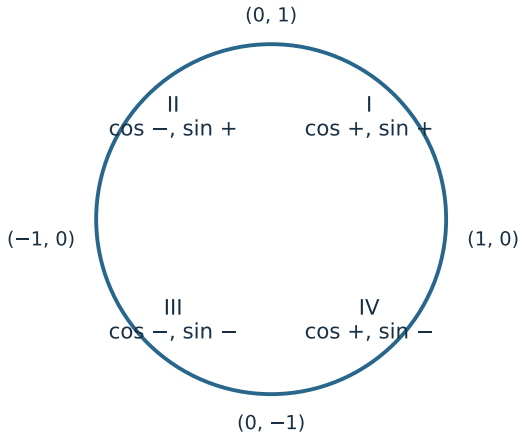
The axis points at 0° and 90° come directly from the circle. They do not require a right triangle with a zero-length leg.

Check 9

Why is $\sin 60°$ greater than $\sin 30°$? Check that the two coordinates at 45° satisfy the unit-circle equation.

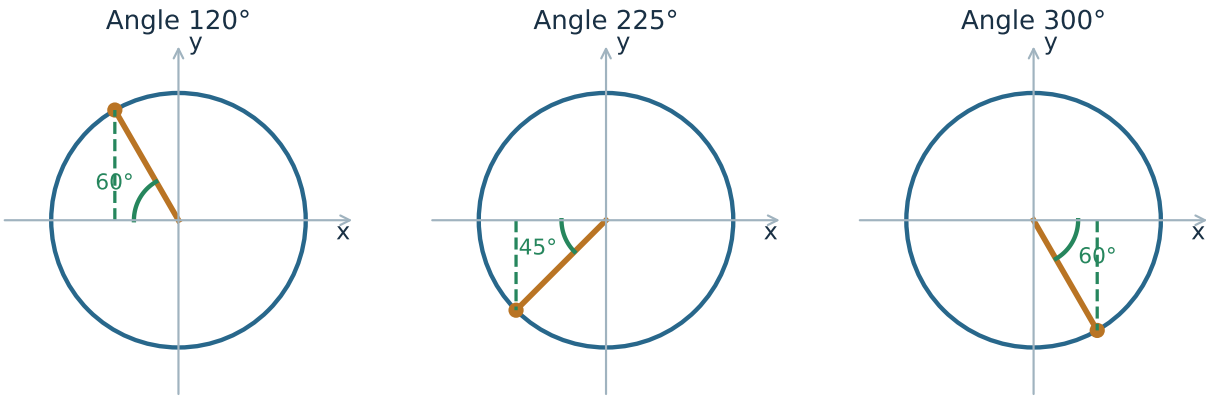
1.10 Use reference angles and quadrant signs

The axes divide the plane into four open regions called *quadrants*. Number them counterclockwise, beginning with the upper-right region. A point on an axis belongs to none of the open quadrants.



The signs follow from left/right and above/below. They are coordinate signs, not an extra rule to memorize.

For a terminal ray away from the axes, the *reference angle* is the acute angle between that ray and the horizontal axis. "Acute" means greater than $0°$ and less than $90°$. Always measure it to the horizontal axis, which may mean the negative x-axis.



Use the reference triangle for magnitudes, then the quadrant for signs.

Find the point at 120°

First locate the angle: $120°$ is in Quadrant II. The reference angle is $180° - 120° = 60°$. At $60°$, the coordinate magnitudes are $1/2$ and $\sqrt{3}/2$. Quadrant II lies left and up, so

$$(\cos 120°, \sin 120°) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

The horizontal coordinate gets the minus sign. The vertical coordinate remains positive. Squaring and adding gives $1/4 + 3/4 = 1$, as it should.

When the original angle includes several turns, first find a coterminal angle in one revolution. Then locate its quadrant and reference angle. For a point on an axis, read the coordinates directly: $(1, 0)$, $(0, 1)$, $(-1, 0)$, or $(0, -1)$. No acute reference triangle is needed.

A useful routine is: locate the ray, find the acute reference angle, determine the two magnitudes, and assign each sign from the picture. Keep the order $(\cos\theta, \sin\theta)$.

Check 10

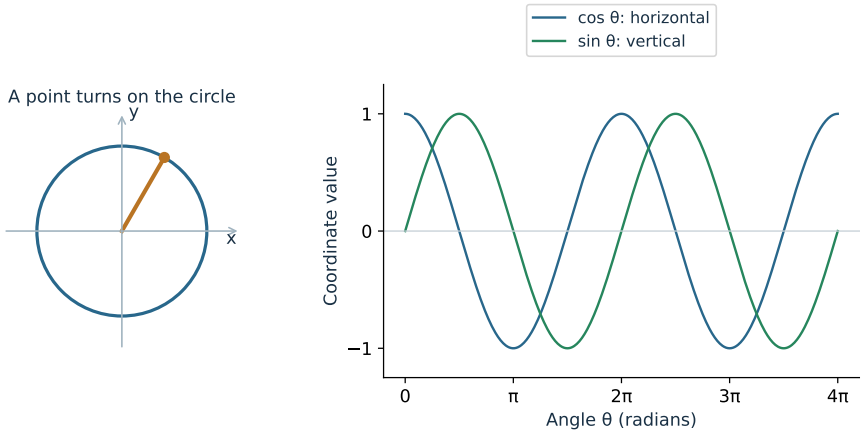
Find the unit-circle point at 300°. Does −60° give the same point? Explain.

1.11 A moving coordinate traces a wave

A point turning around a circle moves in two dimensions. We can also study one coordinate at a time. Make a graph whose horizontal axis is the angle θ, and whose vertical axis is the point's vertical coordinate. That graph is $y = \sin\theta$.

Be careful about the two pictures. On the circle, x and y describe physical position. On the graph of sine, the horizontal axis is the *angle input*; it is not the point's horizontal position.

Angle	Point on the circle	Cosine	Sine
0	$(1, 0)$	1	0
$\pi/2$	$(0, 1)$	0	1
π	$(-1, 0)$	-1	0
$3\pi/2$	$(0, -1)$	0	-1
2π	$(1, 0)$	1	0



One full turn produces one complete cycle of each coordinate graph.

Between these quarter turns, the point moves smoothly, so its coordinates change smoothly. Sine rises from 0 to 1, falls through 0 to -1, and returns to 0. Cosine begins at 1 because the starting point is $(1, 0)$.

A function is *periodic* if its values repeat after a fixed change in input. Adding a complete turn returns the point to exactly the same position:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The period of each is 2π. A half-turn is generally not a period: it moves a point to the opposite side of the circle and reverses both coordinates. For instance, cosine is 1 at 0 and -1 at π.

Explore the connection

In Circle to Wave, stop at each quarter-turn and compare the point's height with the sine graph. Then show two full turns. The playback speed changes how quickly the demonstration runs; the graph against angle keeps the same shape. A graph against time requires a rotation rate, which we introduce next.

Check 11

Why does sine start at 0 while cosine starts at 1? How many complete cycles appear from 0 to 4π?

1.12 Build a model of height over time

To describe a turning wheel over time, connect two steps: time determines an angle, and the angle determines a position. For uniform rotation starting at angle zero,

$$\theta(t) = \omega t.$$

Here t is elapsed time. The notation $\theta(t)$ means the angle at time t. If ω is in rad/s and t is in seconds, their product is in radians.

For a wheel of radius r, centered horizontally at 0 and vertically at height c, with a rider starting at the rightmost point,

$$\boxed{x(t) = r\cos(\omega t), \qquad h(t) = c + r\sin(\omega t)}.$$

The sine term is vertical displacement from the center. Adding c gives height above ground. The radius stretches the unit-circle coordinate; it does not change the value of sine itself.

A radius-8 Ferris wheel

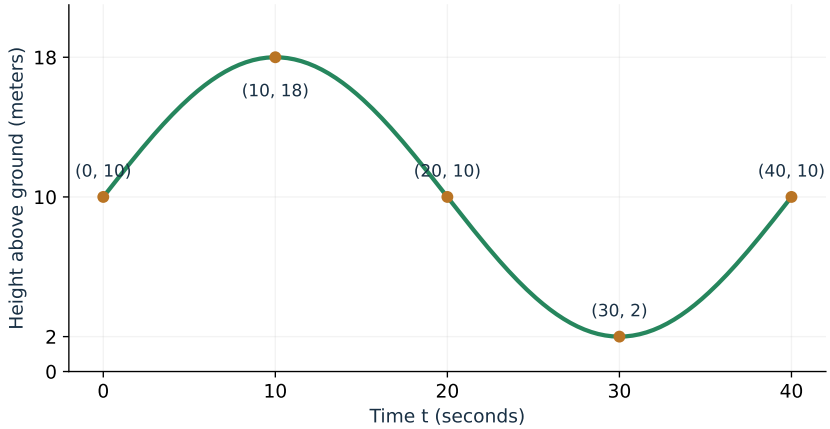
A wheel has radius 8 m, center height 10 m, and period 40 s. The rider starts at the rightmost point, and the wheel turns counterclockwise. First find the angular velocity:

$$\omega = \frac{2\pi}{40} = \frac{\pi}{20} \text{ rad/s}.$$

Next substitute the radius, center height, and angular rate:

$$x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad h(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).$$

At $t = 10$, the angle is $\pi/2$, so the rider is at $(0, 18)$. This is the top: the 10 m center height plus the 8 m radius. At $t = 30$, the angle is $3\pi/2$, so the height is $10 - 8 = 2$ m.



The horizontal axis now measures seconds. The physical assumptions determine where the cycle begins and how long it takes.

Check a model against the situation before trusting a calculated value. The initial height should be 10 m, the height should stay between 2 and 18 m, and one full cycle should take 40 seconds. A negative height would signal a problem with this physical setup.

If a rider begins at another angle θ_0, use $\theta(t) = \theta_0 + \omega t$ inside the coordinate formulas. For clockwise rotation, use a negative ω. Starting position and direction matter; the radius and period alone do not determine a unique height formula.

This is the connection we will develop further: circular motion gives coordinate functions, and those functions let us model repeating behavior.

Check 12

In the example, what is the height at 20 seconds? If the radius became 12 m but the center stayed 10 m high, what would the model predict at the bottom?

1.13 Choose a relationship

Question	Relationship	What to check
How much turning?	$180^\circ = \pi$ rad	Keep the sign; match the unit.
Same terminal ray?	$\theta + 2\pi k$, $k \in \mathbb{Z}$	Use full turns, not half-turns.
How far along the rim?	$d = r\lvert\theta\rvert$	Radians; one direction without reversing.
How fast is the turning?	$\omega = \Delta\theta/\Delta t$	Signed rate, such as rad/s.
How fast is the rim point?	$v = r\lvert\omega\rvert$	Distance/time, such as m/s.
Where is the unit-circle point?	$(\cos\theta, \sin\theta)$	Horizontal first; check signs.
Where is the radius-r point?	$(r\cos\theta, r\sin\theta)$	Center at the origin.
Height on a wheel?	$h = c + r\sin(\omega t)$	Rightmost start, constant rotation rate.

1.14 Practice with explanation

Attempt each problem before consulting the explanations.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? Justify the result using coterminal angles.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.15 Review exercises

1. Convert -135° and 300° to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.
11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.
12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.
13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

1.16 Explanations for the checks

1. The distance is $\sqrt{(-3)^2 + 4^2} = 5$. The normalized point is $(-3/5, 4/5)$. It stays left of the vertical axis and above the horizontal axis.

2. $-90°$ is one quarter-turn clockwise and ends on the negative y-axis. $450°$ is one full turn plus a quarter-turn counterclockwise and ends on the positive y-axis.

3. The angle is $9/6 = 1.5$ radians. Doubling the radius doubles the arc, so the new arc is 18 cm. The angle remains 1.5 radians.

4. $72(\pi/180) = 2\pi/5$ radians, one fifth of a turn. $(7\pi/5)(180/\pi) = 252°$, seven tenths of a turn.

5. $870° - 2(360°) = 150°$. We set aside two full turns. The original rotation was two turns plus $150°$.

6. The angle is $-\pi$ radians. Distance is $2|-\pi| = 2\pi$ m. The endpoints are opposite ends of a diameter, so the displacement has magnitude 4 m.

7. Angular speed is $2\pi/20 = \pi/10$ rad/s. Tangential speed is $2(\pi/10) = \pi/5$ m/s. The units distinguish turning rate from distance rate.

8. Multiply both coordinates by 10: $(-6, -8)$. Its distance from the origin is $\sqrt{36 + 64} = 10$, and both signs remain negative.

9. The point at $60°$ is higher: $\sqrt{3}/2 > 1/2$. At $45°$, $(\sqrt{2}/2)^2 + (\sqrt{2}/2)^2 = 1/2 + 1/2 = 1$.

10. The reference angle is $60°$, and Quadrant IV is right and down. The point is $(1/2, -\sqrt{3}/2)$. Since $-60° + 360° = 300°$, the angles are coterminal and give the same point.

11. The starting point is $(1, 0)$, so the horizontal coordinate is 1 and the vertical coordinate is 0. The interval from 0 to 4π contains two full turns and therefore two cycles.

12. At 20 seconds, $\theta = \pi$, so $h = 10$ m. A radius of 12 m with the same center would predict a bottom height of $10 - 12 = -2$ m. A wheel meant to stay above ground would need a higher center or a different physical setup.

1.17 Practice solutions

1. All three terminal rays coincide. $-120° + 360° = 240°$, and $600° - 360° = 240°$. They point down and left, although the signed rotations differ.

2. $225°(\pi/180°) = 5\pi/4$ radians, in Quadrant III. $(-\pi/3)(180°/\pi) = -60°$, in Quadrant IV.

3. One revolution is 2π radians, so angular speed is $2\pi/40 = \pi/20$ rad/s. Tangential speed is $12(\pi/20) = 3\pi/5$ m/s, about 1.885 m/s.

4. Convert $135° = 3\pi/4$ radians. Arc length is $0.6(3\pi/4) = 9\pi/20$ m. Average angular speed is $(3\pi/4)/3 = \pi/4$ rad/s. Average tangential speed is $(9\pi/20)/3 = 3\pi/20$ m/s, assuming the turn proceeds in one direction.

5. At $150°$, the point is $(-\sqrt{3}/2, 1/2)$. At $330°$, it is $(\sqrt{3}/2, -1/2)$. Corresponding coordinates have equal absolute values, and both signs are opposite.

6. The direction is not specified, so state an assumption. For counterclockwise rotation, $\omega = 2\pi/30 = \pi/15$, $x(t) = 6\cos(\pi t/15)$, and $h(t) = 7 + 6\sin(\pi t/15)$. At 7.5 s the angle is $\pi/2$, so the height is 13 m. Clockwise rotation would instead put the rider at height 1 m at that time.

1.18 Review solutions

1. $-135° = -3\pi/4$ radians; $300° = 5\pi/3$ radians. Multiply each degree value by $\pi/180$.

2. $11\pi/6 = 330°$; $-5\pi/4 = -225°$. Multiply each radian value by $180/\pi$.

3. $19\pi/6 - 2\pi = 19\pi/6 - 12\pi/6 = 7\pi/6$, which is in $[0, 2\pi)$.

4. $s = 8(5\pi/12) = 10\pi/3$ cm. The angle is already in radians.

5. $33\frac{1}{3} = 100/3$ rev/min. Multiply by 2π rad/rev and divide by 60 s/min to get $10\pi/9$ rad/s.

6. They are coordinates on a radius-1 circle centered at the origin. No such point can be farther than 1 unit horizontally or vertically from the origin.

7. At $2\pi/3$: $(-1/2, \sqrt{3}/2)$. At $5\pi/4$: $(-\sqrt{2}/2, -\sqrt{2}/2)$. At $11\pi/6$: $(\sqrt{3}/2, -1/2)$. Use the reference angle for magnitudes and the quadrant for signs.

8. $x = 10\cos\theta$, $y = 12 + 10\sin\theta$. The center adds 12 to the vertical coordinate; the radius multiplies both unit-circle coordinates.

9. $-7\pi/6 + 2\pi = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$, in Quadrant II.

10. In 120 seconds the wheel turns through 18π radians, or 3240°. Divide each by 120: the angular speed is $3\pi/20$ rad/s, or 27°/s.

11. $75° = 5\pi/12$ radians, so the arc is $4(5\pi/12) = 5\pi/3$ m, approximately 5.24 m.

12. The point is $(0, -1)$, on the negative y-axis. Open quadrants exclude their boundary axes, so it belongs to none of them.

13. The radius would increase from 8 to 12 m, increasing the amplitude of each coordinate function to 12. The period remains 40 s and the center height remains 10 m. The predicted height range becomes $[-2, 22]$ m; a wheel staying above ground would require a higher center.

2 The Six Trigonometric Functions

The six trigonometric functions are defined by coordinates on the unit circle and ratios of those coordinates. Their relationships determine their signs, domains, and graphs.

Let $P = (x, y)$ be the point at angle θ on the unit circle. Its coordinates are $x = \cos\theta$ and $y = \sin\theta$. The remaining four functions are formed from quotients and reciprocals of these coordinates.
Learning objectives. Define the six functions on the unit circle, calculate exact values in each quadrant, determine their domains, identify characteristic graph features, and apply trigonometric ratios to measurement problems.

2.1 From a point to six functions

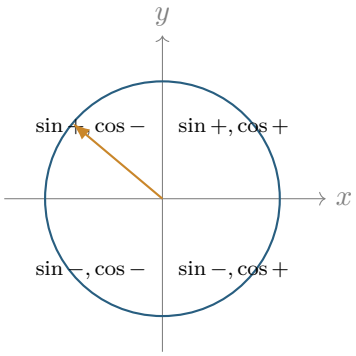
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

Thus the six functions encode related properties of a single point.

Definition 2.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 2.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

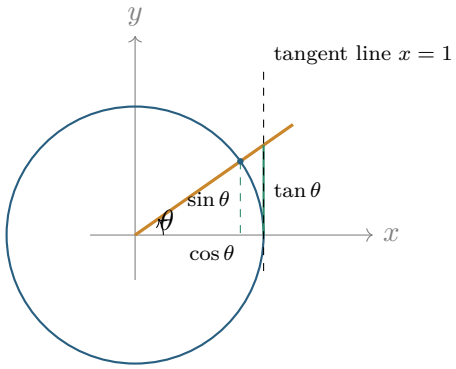


2.2 Tangent as a slope

For $\cos\theta \neq 0$, the line through the origin and $P = (\cos\theta, \sin\theta)$ has slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

Tangent therefore gives the slope of the terminal line. Its magnitude increases without bound as the line approaches a vertical position, at which the slope is undefined.



The terminal line intersects $x = 1$ at $(1, \tan\theta)$, whenever $\cos\theta \neq 0$. Hence tangent also has a geometric interpretation as a directed length on a line tangent to the unit circle.

Example 2.2. At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

This is also the slope of a line rising one unit for every $\sqrt{3}$ units of horizontal displacement.

2.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 2.3. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

The signs of the six functions follow directly from the signs of sine and cosine.

	sin	cos	tan	csc	sec	cot
Quadrant I	+	+	+	+	+	+
Quadrant II	+	−	−	+	−	−
Quadrant III	−	−	+	−	−	+
Quadrant IV	−	+	−	−	+	−

Try it. The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

2.4 Domains and undefined values

Division by zero is undefined. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. Undefined values must be distinguished from zeros of a function.

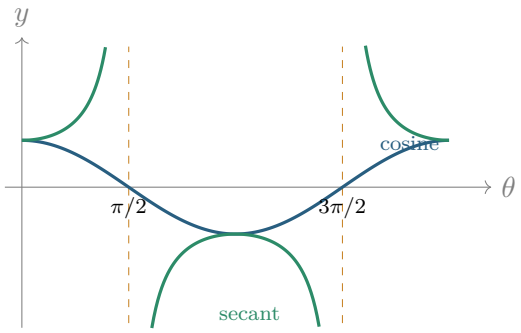
Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

2.5 Zeros and vertical asymptotes

Near zeros of their denominators, trigonometric quotients may become arbitrarily large in magnitude. This behavior accounts for their vertical asymptotes.

For example, as θ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

The line $\theta = \pi/2$ is a vertical asymptote of both tangent and secant.



Only those portions of the secant graph within the displayed vertical range are shown. Its branches are separated by vertical asymptotes.

Try it. Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

2.6 Right-triangle ratios

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit-circle definitions extend the right-triangle ratios to arbitrary real angles.

Example 2.4. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

Writing o, a, and h for opposite side, adjacent side, and hypotenuse, respectively, gives

	opposite and hypotenuse	adjacent and hypotenuse	opposite and adjacent
direct	$\sin\theta = o/h$	$\cos\theta = a/h$	$\tan\theta = o/a$
reciprocal	$\csc\theta = h/o$	$\sec\theta = h/a$	$\cot\theta = a/o$

The six ratios describe the same triangle and agree with the unit-circle definitions for acute angles.

Example 2.5. A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If θ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

Each quotient expresses a relation between lengths of the same triangle.

2.7 Exact values at special angles

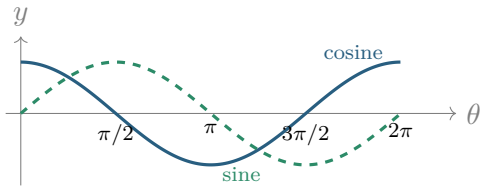
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.



2.8 Graphs of the trigonometric functions

On a trigonometric graph, the independent variable is an angle and the dependent variable is a coordinate or ratio at that angle. Such a graph is distinct from the circular trajectory of the associated point.

Sine and cosine remain between -1 and 1. Both repeat after a full rotation, so their fundamental period is 2π radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is π radians, or 180°. Secant and cosecant share the 2π period of their reciprocal partners.



The zeros, excluded inputs, and fundamental period determine important features of a trigonometric graph.

Example 2.6. Rotation by π changes the signs of both unit-circle coordinates. Thus

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

Therefore tangent has period π on its domain.

The accompanying Six Trigonometric Graphs and Reciprocal Trigonometric Graphs demonstrations illustrate these functions and their asymptotes.

Try it. Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

2.9 Applications: orientation and measurement

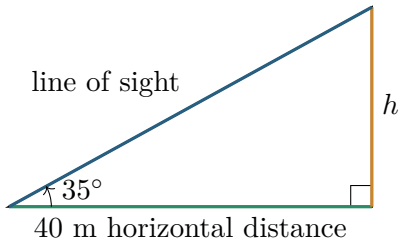
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

2.10 Measurement with trigonometric ratios

The appropriate trigonometric ratio depends on the available lengths. Tangent relates height to horizontal distance; sine relates height to line-of-sight distance.



Worked example: height measurement. An observer is 20 meters from the base of a tree and measures an angle of elevation of 40° to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The observer's eye height must be added to determine the height above ground.

Try it. A rescue drone is observed at an elevation angle of 25° from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

2.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?
5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.
6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.
7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.
9. Find the exact values of all six trig functions at $11\pi/6$.
10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.
11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.
12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.
13. If $\cos\theta = -12/13$ and θ lies in quadrant II, determine all other trig values exactly.
14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.
15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.

3 Periodic Functions and Their Graphs

Periodic functions describe quantities that repeat as their input increases. Trigonometric functions provide fundamental examples: their periodicity arises from rotation, and their graphs can be transformed to represent a wide range of oscillatory processes.

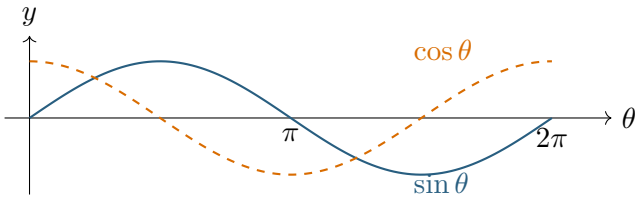
Learning objectives. Identify periods, amplitudes, midlines, and phase shifts; construct transformed sine and cosine graphs; distinguish angular frequency from frequency; and determine sinusoidal models from geometric or measured information.

3.1 Circular motion and periodicity

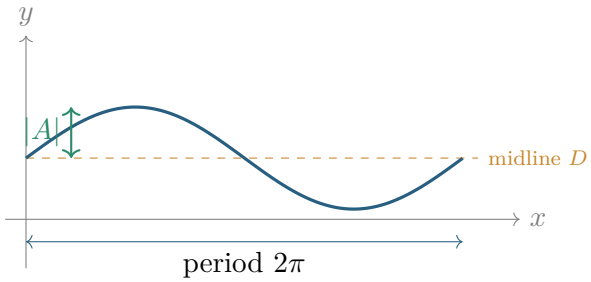
For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

A function f is *periodic* if there exists $T > 0$ such that $f(x + T) = f(x)$ for every x in its domain. When a least positive such T exists, it is the *fundamental period*. The fundamental periods of sine and cosine are both 2π. Moreover, $\sin 0 = 0$ and $\cos 0 = 1$.



3.2 Amplitude and the midline



3.3 Vertical transformations

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 3.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

For a sinusoid with maximum M and minimum m, the amplitude and midline are

$$|A| = \frac{M - m}{2}, \qquad D = \frac{M + m}{2}.$$

These relations determine the vertical parameters without knowing the phase or period.

Example 3.2. A periodically varying temperature has maximum 26°C and minimum 14°C. Its midline is 20°C and its amplitude is 6°C. These values do not determine the period or phase.

Try it. A sinusoidal signal ranges from -7 to 5. Determine its amplitude, midline, and range, and sketch one possible cycle.

3.4 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 3.3. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

A phase change of 2π radians corresponds to one cycle. If the argument of a sinusoidal function is $\omega t + \phi$, then its period is $T = 2\pi/|\omega|$, and its frequency is

$$f = \frac{1}{T} = \frac{|\omega|}{2\pi}.$$

Frequency is measured in cycles per unit time, while angular frequency is measured in radians per unit time. These units describe the same repetition rate in different ways.

Example 3.4. A mechanism rotates at 90 revolutions per minute. Its frequency is 1.5 cycles per second, and its angular speed is $2\pi(1.5) = 3\pi$ radians per second. Its period is 2/3 second.

3.5 Phase shift

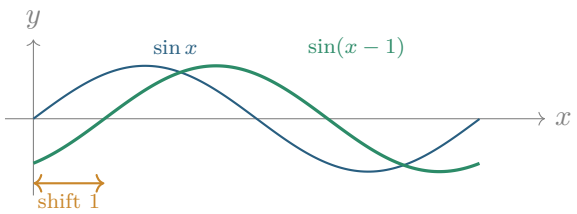
A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 3.5. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

The parameterization assumes $B \neq 0$. In factored form, the phase shift is C. In the equivalent form $A\sin(Bx + \phi) + D$, the same shift is $C = -\phi/B$. The sign of B may be taken positive in a standard sine model by modifying the sign of A.



3.6 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

For $y = A\sin(B(x-C)) + D$ with $B > 0$, write $T = 2\pi/B$. One cycle is determined by the following five points:

x	C	$C+T/4$	$C+T/2$	$C+3T/4$	$C+T$
y	D	$D+A$	D	$D-A$	D

For cosine, the corresponding outputs are $D+A, D, D-A, D, D+A$. These points determine the positions of maxima, minima, and midline crossings; the smooth sinusoidal curve passes through them.

Example 3.6. For $y = -2\cos(\pi(x-1)) + 3$, the period is $T = 2$. Beginning at $x = 1$, the five landmark points are

$$(1,1), \quad (3/2,3), \quad (2,5), \quad (5/2,3), \quad (3,1).$$

The negative coefficient places the minimum at the start of the cycle.

3.7 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

3.8 Constructing a model from measurements

Suppose a buoy reaches a maximum height of 5 meters and a minimum height of 1 meter. Successive maxima are separated by 8 seconds. Taking $t = 0$ at a maximum yields the model

$$h(t) = 3 + 2\cos\left(\frac{\pi t}{4}\right).$$

The midline is 3 meters, the amplitude is 2 meters, and the angular frequency is $\pi/4$ radians per second. If the time origin were chosen at an upward midline crossing, a sine model would be more convenient. The physical motion is unchanged by this choice of representation.

Try it. A tide measurement varies between 0.8 m and 3.6 m and repeats every 12 hours. Write a cosine model assuming that the maximum occurs at $t = 2$ hours. State the meaning and units of every parameter.

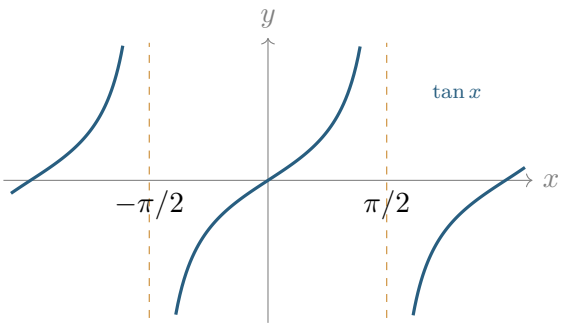
3.9 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

Because $\tan x = \sin x/\cos x$, its zeros occur at $x = k\pi$ and its vertical asymptotes at $x = \pi/2 + k\pi$. Similarly, $\cot x = \cos x/\sin x$ is undefined at integer multiples of π. The graphs of secant and cosecant lie outside the strip $-1 < y < 1$ because their absolute values are at least 1 where defined.



3.10 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.
2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.
3. Give five landmark points for $y = 2\sin(x - \pi/2)$.
4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.
5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.
6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.
8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.
9. Determine a sinusoidal model with maximum 8, minimum -2, period 10, and maximum at $x = 3$.
10. Explain why $A\sin(B(x-C))+D$ is constant when $A = 0$, and why the usual fundamental-period formula does not apply.
11. A motor oscillates at 2.5 cycles per second. Determine its angular frequency and period.
12. Give the domain of $y = \tan(3x)$ and identify two consecutive vertical asymptotes.
13. Determine the amplitude and midline from a sinusoid with maximum 12.4 and minimum 7.6.

4 Inverse Trigonometry and Identities

Inverse trigonometric functions recover principal angles from trigonometric values. Their definitions require restrictions of the original domains. Trigonometric identities, in turn, express relationships among functions of the same angle and provide methods for simplifying and evaluating expressions.

Learning objectives. Determine the domains and ranges of the principal inverse trigonometric functions; evaluate and compose inverse and trigonometric functions; recover exact values from right triangles; prove identities using fundamental relations; and derive and apply angle-addition and double-angle formulas.

4.1 Inverse functions and principal branches

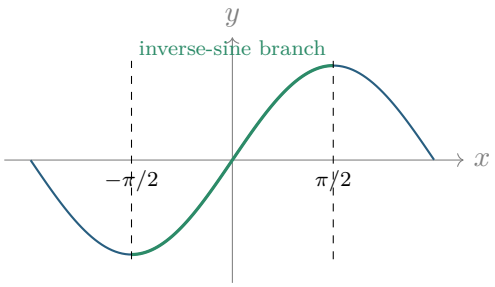
The notation $\sin^{-1} x$ commonly denotes $\arcsin x$, the inverse sine function. It must not be confused with $\csc x = 1/\sin x$. The inverse returns an angle from a prescribed trigonometric value, whereas the reciprocal returns a multiplicative inverse of that value.

A periodic function is not one-to-one on its full domain and therefore does not admit a global inverse function. To define the principal inverse sine, sine is restricted to $[-\pi/2, \pi/2]$, where it is strictly increasing and maps onto $[-1, 1]$. Similarly, cosine is restricted to $[0, \pi]$ and tangent to $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

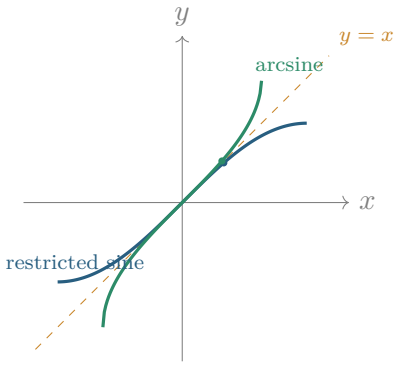
Example 4.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 4.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.



4.2 Reflection across the diagonal

The graphs of an invertible function and its inverse are reflections across the line $y = x$: if (a, b) lies on the graph of f, then (b, a) lies on the graph of f^{-1}. For inverse sine, only the restricted sine branch is reflected, rather than the entire periodic sine graph.



The reflected domains and ranges exchange roles. In particular, $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ whereas the restricted sine maps $[-\pi/2, \pi/2]$ onto $[-1,1]$.

Try it. List three points on $y = \cos x$ for $0 \le x \le \pi$, and interchange their coordinates to obtain points on $y = \arccos x$.

4.3 Compositions and triangle geometry

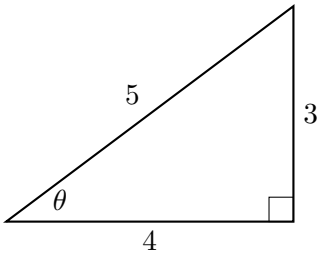
The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

For the inverse compositions, the domain restrictions determine which cancellations are valid:

$$\arcsin(\sin x) = x \quad \text{for } -\pi/2 \le x \le \pi/2, \qquad \sin(\arcsin u) = u \quad \text{for } -1 \le u \le 1.$$

The first identity does not hold for arbitrary real x, since the arcsine output is restricted to its principal range. Analogous statements hold for cosine and tangent on their chosen branches.



$\theta = \arctan(3/4)$

$\sin\theta = 3/5$

Example 4.3. To evaluate $\cos(\arcsin(-5/13))$, let $\theta = \arcsin(-5/13)$. Since $\theta \in [-\pi/2, \pi/2]$, its cosine is nonnegative. Thus

$$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}.$$

The principal range determines the sign of the square root.

Try it. Evaluate $\sin(\arccos(-3/5))$ exactly. Explain the sign of the result using the principal range of arccosine.

4.4 Identities and their domains

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1+\tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1+\cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 4.4. For $\sin x \neq 0$,

$$\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

4.5 Symmetry identities

The unit circle is symmetric about both coordinate axes. Reflection across the horizontal axis changes the sign of the vertical coordinate without changing the horizontal coordinate. Consequently,

$$\cos(-x) = \cos x, \qquad \sin(-x) = -\sin x.$$

The cosine function is *even*, while sine is *odd*. Taking quotients gives $\tan(-x) = -\tan x$ wherever tangent is defined.

A half-turn changes both coordinate signs:

$$\cos(x+\pi) = -\cos x, \qquad \sin(x+\pi) = -\sin x.$$

These relations explain the π-periodicity of tangent.

Try it. Use the unit-circle symmetries to evaluate $\sin(-7\pi/6)$ and $\cos(-7\pi/6)$ without a calculator.

4.6 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 4.5. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6}+\sqrt{2})/4$.

4.7 A geometric derivation of the addition formulas

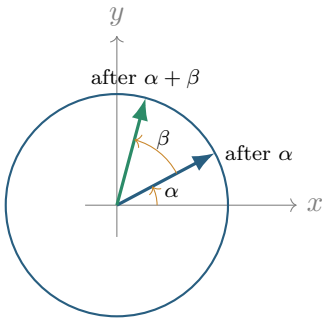
Rotation through an angle α sends a point (x, y) to

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In particular, apply this rotation to the unit-circle point $(\cos\beta, \sin\beta)$. The result must be the point $(\cos(\alpha+\beta), \sin(\alpha+\beta))$, since successive rotations add their angles. Matrix multiplication gives

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Thus the sum formulas express the geometry of composing rotations, rather than independent algebraic rules.



4.8 Double-angle and half-angle consequences

Setting $\alpha = \beta = x$ gives

$$\sin 2x = 2\sin x \cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Rearranging the cosine identity produces the power-reduction relations

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These expressions are particularly useful when simplifying squared trigonometric functions.

Example 4.6. The exact value of $\cos 15^\circ$ follows by using $15^\circ = 45^\circ - 30^\circ$:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Try it. Use the addition identities to find $\sin 105^\circ$ exactly. Then verify the sign from the quadrant containing 105°.

4.9 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9^\circ$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

4.10 Recovering angles in measurement problems

The inverse function identifies an angle from a ratio, provided the physical geometry determines an appropriate branch. If a right triangle has adjacent side 40 and opposite side 30, its acute angle satisfies

$$\theta = \arctan(30/40) = \arctan(3/4) \approx 36.87^\circ.$$

The same geometry gives the opposite side directly from $h = 40\tan\theta$, when the angle is already known. In a measurement context, the acute-angle restriction is supplied by the geometry of the triangle.

4.11 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.

2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.

3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75^\circ)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.
9. State the domain and range of $\arccos x$, and explain why the restriction $[0, \pi]$ is appropriate.
10. Evaluate $\arccos(\cos(5\pi/3))$ exactly.
11. Find $\tan(\arccos(4/5))$ exactly using a right triangle.
12. Prove $\sin(-x) = -\sin x$ geometrically and use it to compute $\sin(-\pi/3)$.
13. Verify the cosine difference identity by replacing β with $-\beta$ in the cosine sum identity.
14. Prove $\cos^2 x = (1 + \cos 2x)/2$ on all of $\mathbb{R}$.
15. Find $\sin 15^\circ$ exactly and state its sign.
16. Explain why $\arctan(\tan(3\pi/4))$ is not $3\pi/4$.

5 Trigonometric Equations and Applications

Central question. When a trigonometric function has a prescribed value, how can we find *all* the angles that produce it? How can we use angles to solve triangles that are not right triangles?

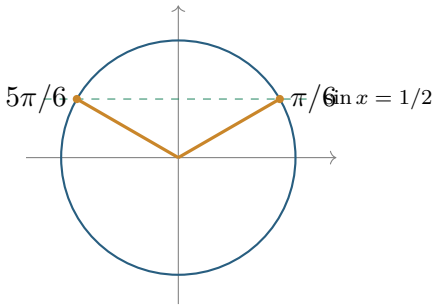
5.1 A first equation on the unit circle

Consider $\sin x = 1/2$. The unit circle has two points whose vertical coordinate is $1/2$, at $x = \pi/6$ and $x = 5\pi/6$. Every full revolution reproduces these points:

$$\boxed{x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

An inverse trigonometric function gives one principal angle, not necessarily every solution. The unit circle and periodicity give the complete answer.

Example 5.1. Solve $\cos x = -\sqrt{2}/2$ for $0 \le x < 2\pi$. The solutions are $3\pi/4$ and $5\pi/4$.



5.2 Equations with transformations

To solve $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$, first solve for $u = 2x$ on $[0, 4\pi)$. We obtain $u = \pi/3, 2\pi/3, 7\pi/3, 8\pi/3$ and therefore

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}.$$

A changed frequency changes how many full cycles fit into the requested interval.

Example 5.2. Solve $2\cos x - 1 = 0$ on $[0, 2\pi)$. Isolating cosine gives $\cos x = 1/2$, hence $x = \pi/3$ or $5\pi/3$.

5.3 Factoring and checking domains

An equation may contain more than one trig function. Algebraic techniques still work:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

On $[0, 2\pi)$ we obtain $x = 0, \pi, \pi/6, 5\pi/6$. Dividing by $\sin x$ at the outset would have lost two solutions.

5.4 Trigonometric identities in equations

For example, $2\cos^2 x - 1 = 0$ becomes $\cos^2 x = 1/2$, hence $\cos x = \pm\sqrt{2}/2$. There are four solutions on $[0, 2\pi)$. Using identities can make equations simpler, but each transformation should preserve the original domain.

5.5 Solving triangles: notation

Write the vertices of a triangle as A, B, C, and the opposite side lengths as a, b, c. Its angles satisfy $A + B + C = \pi$. For right triangles, the six trig ratios are enough. For oblique triangles, two laws extend our tools.

Definition 5.1 (Law of Sines)**.** For every nondegenerate triangle,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.}$$

Example 5.3. A triangle has $A = 40°$, $B = 65°$, and $a = 9$ cm. Since $C = 75°$, the Law of Sines gives $b = 9 \sin 65° / \sin 40° \approx 12.69$ cm.

5.6 The ambiguous SSA case

Given two sides and a nonincluded angle (SSA), the Law of Sines may produce zero, one, or two triangles. If $\sin B = s$, both $B = \arcsin s$ and $B = \pi - \arcsin s$ must be considered, and each candidate must fit with the known angle A in a triangle.

Example 5.4. Suppose $A = 30°$, $a = 5$ and $b = 8$. Then $\sin B = 8 \sin 30°/5 = 0.8$. Both $B \approx 53.13°$ and $B \approx 126.87°$ are consistent with $A + B < 180°$, so two distinct triangles exist.

Definition 5.2 (Law of Cosines)**.** For a triangle with sides a, b, c opposite angles A, B, C,

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C.}$$

When $C = 90°$ this becomes the Pythagorean theorem. The cosine term measures how the included angle modifies the length of the opposite side.

Example 5.5. Two sides of a triangle are 7 m and 10 m, with included angle $60°$. The opposite side satisfies $c^2 = 49 + 100 - 140(1/2) = 79$, so $c = \sqrt{79}$ m.

5.7 Applied modeling and interpretation

Two sensors observe a distant object from different locations. The baseline and their sight angles make a triangle. The Law of Sines can recover an inaccessible distance if we know a side and two angles. Always include the diagram, angle units, and a reasonableness check.

5.8 Exercises

1. Solve $\sin x = -\sqrt{3}/2$ for $0 \le x < 2\pi$.
2. Solve $2 \sin(3x) = 1$ for $0 \le x < 2\pi$; find every solution.
3. Solve $2 \cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. A triangle has $A = 35°$, $B = 80°$ and $a = 12$. Find C and b.
6. For $A = 30°$, $a = 5$, $b = 8$, calculate both possible values of C in the SSA case.
7. Find the third side of a triangle when two sides have length 6 and 9 and their included angle is $120°$.

8. Explain why the Law of Cosines becomes the Pythagorean theorem for right triangles.

6 Vectors and Geometric Applications

Central question. How do we describe a quantity that has both a size and a direction?

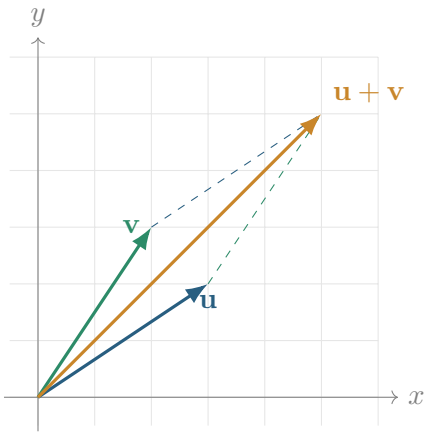
6.1 From coordinates to directed displacements

In the first half of the course we used $(\cos\theta, \sin\theta)$ to locate a point on a circle. A vector uses the same coordinates to describe a *displacement*: how far to move horizontally and vertically. Write $\mathbf{v} = \langle a, b\rangle$. Its magnitude is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}}.$$

Vectors with the same components represent the same displacement even when drawn at different starting points.

Example 6.1. A walker moves 3 km east and 4 km north. The resulting vector is $\langle 3, 4\rangle$ km and its magnitude is 5 km. The route traveled is 7 km, while the displacement is only 5 km.



6.2 Vector arithmetic

Addition combines displacements:

$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle.$$

Scalar multiplication changes their length, and a negative scalar reverses direction:

$$k\langle a, b\rangle = \langle ka, kb\rangle.$$

Subtracting vectors compares locations. If $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then $\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2\rangle$.

Example 6.2. For $\mathbf{u} = \langle 2, -1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$, we have $\mathbf{u} + \mathbf{v} = \langle -1, 3\rangle$ and $2\mathbf{u} - \mathbf{v} = \langle 7, -6\rangle$.

6.3 Direction and trigonometry

A unit vector has magnitude 1. The unit vector pointing at angle θ is

$$\mathbf{e}_\theta = \langle \cos\theta, \sin\theta\rangle.$$

A vector of magnitude r in that direction is

$$\boxed{\mathbf{v} = r\langle \cos\theta, \sin\theta\rangle}.$$

Conversely, for a nonzero vector $\langle a, b\rangle$, the angle must be chosen in the appropriate quadrant. An inverse tangent applied only to b/a may miss that quadrant.

Example 6.3. A force of 20 N directed 30° above the horizontal has components $\langle 10\sqrt{3}, 10\rangle$ N.

6.4 The dot product measures alignment

Define

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2.$$

The geometric interpretation is

$$\boxed{\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\phi},$$

where ϕ is the angle between nonzero vectors. Positive dot product means an acute angle; negative means an obtuse angle; zero means perpendicular vectors.

Example 6.4. With $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, the dot product is $12 - 12 = 0$. Both vectors are nonzero, so they are perpendicular.

6.5 Projection and useful components

The scalar component of $\mathbf{u}$ in the direction of a nonzero vector $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

Projection is the part of one vector aligned with another. In physics, the work done by a constant force $\mathbf{F}$ across displacement $\mathbf{d}$ is $W = \mathbf{F}\cdot\mathbf{d}$, measured in joules when force is in newtons and displacement in meters.

Example 6.5. A force $\langle 10, 5\rangle$ N acts during displacement $\langle 3, 0\rangle$ m. The work is 30 J: only the horizontal force component contributes.

6.6 A geometric connection to the Law of Cosines

For $\mathbf{u}, \mathbf{v}$ we have

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\phi.$$

Thus the Law of Cosines from Module 5 is also a theorem about vector differences.

6.7 Exercises

1. Find the vector from $P = (-2, 3)$ to $Q = (5, -1)$ and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at angle 150°.
4. Find a unit vector in the direction of $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.

7. Find $\text{proj}_{\mathbf{v}}\,\mathbf{u}$ when $\mathbf{u} = \langle 4, 3\rangle$ and $\mathbf{v} = \langle 1, 0\rangle$.

8. A force of 50 N acts 60° from the direction of a 4 m displacement. Calculate the work done.

7 Polar Coordinates and Complex Numbers

Central question. What happens when we describe a point by its distance and direction instead of its horizontal and vertical coordinates?

7.1 Two descriptions of the same point

Cartesian coordinates specify (x, y). Polar coordinates specify (r, θ), where r is the directed radial distance and θ is an angle from the positive horizontal axis:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta.}$$

For $r \geq 0$, the distance is $r = \sqrt{x^2 + y^2}$. The angle must be chosen using the signs of both coordinates, not just $\arctan(y/x)$.

Example 7.1. The polar point $(4, 5\pi/6)$ corresponds to $(x, y) = (-2\sqrt{3}, 2)$. Its distance from the origin is 4, and it lies in Quadrant II.

Try it. Convert $(r, \theta) = (6, -\pi/3)$ to Cartesian coordinates. Then describe its direction using an angle in $[0, 2\pi)$.

7.2 Polar coordinates are not unique

Angles that differ by 2π represent the same direction, so (r, θ) and $(r, \theta + 2\pi k)$ describe the same point. A negative radius reverses the ray:

$$(r, \theta) = (-r, \theta + \pi).$$

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ are three descriptions of the same point. At the origin $r = 0$, every angle describes the same location.

7.3 Complex numbers as plane vectors

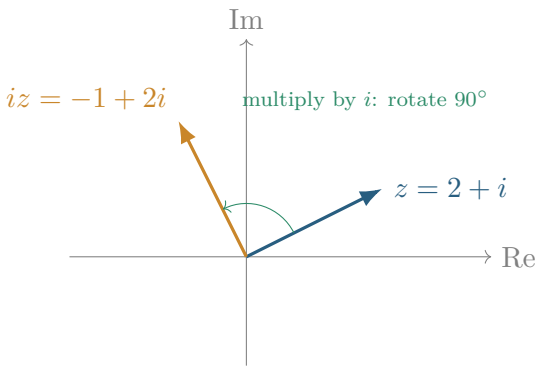
Introduce i satisfying $i^2 = -1$. A complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane. Its modulus and argument are

$$|z| = \sqrt{a^2 + b^2}, \qquad \arg z = \theta \quad \text{when } z = |z|(\cos\theta + i\sin\theta).$$

The argument of $z = 0$ is undefined. As with polar angles, nonzero arguments are only unique after choosing a branch, such as $(-\pi, \pi]$.

Example 7.2. For $z = -\sqrt{3} + i$, the modulus is 2 and one argument is $5\pi/6$. Thus

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$



7.4 Addition and multiplication mean different geometries

Complex addition follows the vector rule:

$$(a+bi)+(c+di)=(a+c)+(b+d)i.$$

Complex multiplication uses $i^2=-1$:

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i.$$

If $z=r(\cos\alpha+i\sin\alpha)$ and $w=s(\cos\beta+i\sin\beta)$, the angle-addition formulas give

$$\boxed{zw=rs(\cos(\alpha+\beta)+i\sin(\alpha+\beta)).}$$

Therefore multiplying complex numbers multiplies distances from the origin and *adds angles*. A unit complex number acts as a rotation. This unifies the unit circle from Module 1 with vector geometry from Module 6.

Example 7.3. Multiplying $2+i$ by i gives $i(2+i)=-1+2i$. The vector $(2,1)$ rotates counterclockwise by 90° to $(-1,2)$, without changing its length.

7.5 De Moivre's theorem and roots

For integer n, repeated multiplication gives

$$\boxed{[r(\cos\theta+i\sin\theta)]^n=r^n(\cos n\theta+i\sin n\theta)}$$

whenever the expression makes sense; for negative n, require $r>0$. For example, $(\cos(\pi/6)+i\sin(\pi/6))^3=i$. To solve $z^n=R(\cos\phi+i\sin\phi)$ with $R>0$, take

$$z_k=R^{1/n}\left(\cos\frac{\phi+2\pi k}{n}+i\sin\frac{\phi+2\pi k}{n}\right),\qquad k=0,\dots,n-1.$$

These n distinct roots are equally spaced around a circle.

Example 7.4. The four fourth roots of 1 are $1,i,-1,-i$. They are separated by a quarter-turn.

7.6 Polar graphs: a different way to draw a curve

In a polar equation such as $r=2\cos\theta$, the radius changes with the angle. Multiply by r: $r^2=2r\cos\theta$, or $x^2+y^2=2x$. Completing the square yields

$$(x-1)^2+y^2=1.$$

Thus a simple polar equation can represent an ordinary circle displaced from the origin.

7.7 Exercises

1. Convert $(5,2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar coordinate representations of $(\sqrt{2},\sqrt{2})$.
3. Write $-1-\sqrt{3}i$ in polar form with argument in $[0,2\pi)$.
4. Calculate $(2+i)(1-3i)$ in rectangular form.

5. Interpret multiplication by $-i$ as a plane transformation.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find all three cube roots of 1 and plot them on the unit circle.
8. Convert the polar curve $r = 4\sin\theta$ into a Cartesian equation and identify the curve.

8 Parametric Equations and Conic Sections

Central question. How can the same curve be described by coordinates, by a moving point, or by a geometric rule?

8.1 A moving point traces a curve

In a parametric description, both coordinates depend on a parameter t:

$$\boxed{x = f(t), \qquad y = g(t).}$$

Think of t as time when a particle moves. The curve records its location, while the parameterization also records direction and timing.

Example 8.1. For $x = 2t + 1$, $y = t - 3$, eliminating $t = (x - 1)/2$ gives $y = (x - 7)/2$. If t increases, the point moves up and right along this line.

8.2 Eliminating the parameter

For $x = 3\cos t$, $y = 2\sin t$, square the normalized equations and add:

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1.}$$

The equation gives the whole ellipse when t ranges over $[0, 2\pi]$. A restricted interval, such as $[0, \pi]$, traces only its upper half. Different parameterizations may have the same image but different speeds or directions.

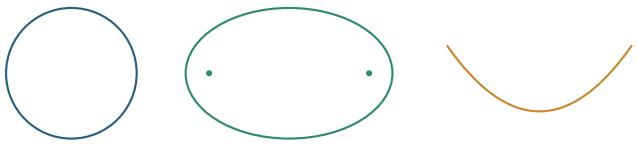
Example 8.2. Both $(x, y) = (\cos t, \sin t)$ and $(x, y) = (\cos 2t, \sin 2t)$ trace the unit circle for $0 \le t \le 2\pi$; the second makes two revolutions.

8.3 Projectile motion as a parametric curve

Under constant downward acceleration g, a projectile launched from (x_0, y_0) at speed v_0 and angle θ satisfies

$$x(t) = x_0 + v_0 \cos\theta\, t, \qquad y(t) = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

When the horizontal velocity is nonzero, eliminating t gives a parabola. The motion model adds information the Cartesian curve alone does not supply, including direction and location at a given time.



circle ellipse with foci parabola

8.4 Conics: circles and ellipses

A *circle* is the set of points at a fixed distance r from a center (h, k):

$$(x-h)^2 + (y-k)^2 = r^2.$$

An *ellipse* can be defined by a constant sum of distances to two foci. In standard form, for $a \geq b > 0$,

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

Its horizontal semiaxis is a and vertical semiaxis is b; when the major axis is horizontal, focal distance obeys $c^2 = a^2 - b^2$.

Example 8.3. For $x^2/25 + y^2/9 = 1$, the vertices are $(\pm 5, 0)$, and the foci are $(\pm 4, 0)$.

8.5 Parabolas and their focus

A *parabola* is the set of points equidistant from a fixed point (focus) and a fixed line (directrix). The equation

$$(x-h)^2 = 4p(y-k)$$

has vertex (h, k), focus $(h, k+p)$, and directrix $y = k - p$. Its axis is vertical; $p > 0$ opens upward and $p < 0$ downward.

Example 8.4. For $x^2 = 8y$, we have $4p = 8$, so the focus is $(0, 2)$ and the directrix is $y = -2$.

8.6 Hyperbolas and differences of distances

A *hyperbola* consists of points for which the absolute difference of distances to two foci is fixed. A horizontal hyperbola has standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

Its vertices are $(h \pm a, k)$, its foci are $(h \pm c, k)$ where $c^2 = a^2 + b^2$, and its asymptotes are $y - k = \pm (b/a)(x-h)$.

Example 8.5. The hyperbola $x^2/9 - y^2/16 = 1$ has vertices $(\pm 3, 0)$, foci $(\pm 5, 0)$, and asymptotes $y = \pm 4x/3$.

8.7 Changing descriptions

A parametric curve may be easier to generate than to recognize from a Cartesian equation. A Cartesian equation reveals symmetry or an axis. A focal definition explains why the curve arises in optics, navigation, and geometry. None of these descriptions replaces the others; each answers a different question.

8.8 Exercises

1. Eliminate t from $x = t + 2$, $y = 3t - 1$ and describe the motion as t increases.

2. Find the Cartesian equation traced by $x = 4\cos t$, $y = 3\sin t$.

3. Explain how $(\cos t, \sin t)$ and $(\cos 3t, \sin 3t)$ differ as parameterizations.

4. A projectile leaves the origin at 20 m/s at 30°, with $g = 9.8$ m/s^2. Write its parametric equations.
5. Identify center and radius of $(x-2)^2 + (y+1)^2 = 16$.
6. Find the foci of $x^2/36 + y^2/20 = 1$.
7. Find the focus and directrix of $(x-1)^2 = 12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4 - y^2/9 = 1$.

9 Sequences and Series

Central question. How do repeated steps produce patterns, and how can we predict both a later value and a cumulative total?

9.1 A sequence is an ordered list

A sequence assigns a number a_n to each positive integer n. For example, $a_n = 3 + 2(n-1)$ produces $3, 5, 7, 9, \ldots$. We distinguish the *term* a_n from a *partial sum* $S_n = a_1 + \cdots + a_n$.

9.2 Arithmetic sequences: repeated addition

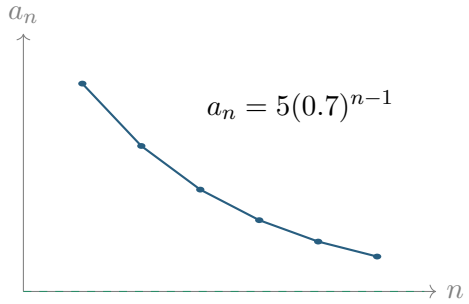
An arithmetic sequence has a fixed common difference d:

$$a_{n+1} = a_n + d, \qquad \boxed{a_n = a_1 + (n-1)d}.$$

Example 9.1. A theater's first row has 12 seats, with two more seats in each successive row. Row 15 has $12 + 14(2) = 40$ seats.

Adding the first and last terms in reverse order gives

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)}.$$



9.3 Geometric sequences: repeated multiplication

A geometric sequence has a fixed common ratio q:

$$a_{n+1} = qa_n, \qquad \boxed{a_n = a_1 q^{n-1}}.$$

When $q > 1$, positive terms grow exponentially. When $0 < q < 1$, they decay toward zero.

Example 9.2. A population of 500 organisms increases by 8 percent each generation. The model is $a_n = 500(1.08)^{n-1}$ if generation 1 is the initial observation.

For $q \neq 1$, multiply the finite sum by q and subtract:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}, \qquad \boxed{S_n = a_1 \frac{1 - q^n}{1 - q}}.$$

If $q = 1$, the sum is na_1. The formulas refer to *finite* sums and work for ratios outside the interval $(0, 1)$ as well.

9.4 Sigma notation

The compact expression $\sum_{k=1}^{n} a_k$ means add the terms for integer indices $k = 1, \dots, n$. For instance,

$$\sum_{k=1}^{4} (2k+1) = 3 + 5 + 7 + 9 = 24.$$

The symbol $\sum$ tells us to add, while its bounds tell us which indices are included.

9.5 Infinite geometric series

If $|q| < 1$, then $q^n \to 0$. The finite geometric sum approaches

$$\boxed{\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}}, \qquad |q| < 1.$$

The condition matters: the series $1 + 2 + 4 + \cdots$ does not approach a finite sum.

Example 9.3. A ball bounces to 60% of its previous height. Released from height 5 m, the total vertical travel after infinitely many idealized bounces is $5+2(5)(0.6+0.6^2+\cdots) = 5+10(0.6/0.4) = 20$ m.

9.6 Recursive rules beyond arithmetic and geometric patterns

Not every sequence is arithmetic or geometric. Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

The first terms are $0, 3, 4.5, 5.25, \dots$. Let $e_n = a_n - 6$ denote the difference from the fixed point. Then $e_{n+1} = e_n/2$, so $e_n = -6(1/2)^{n-1}$ and

$$a_n = 6 - 6(1/2)^{n-1}.$$

The original recursion is not geometric, but its *error from equilibrium* is. This is a useful bridge to feedback and dynamical models.

9.7 Exercises

1. Write the first five terms of $a_n = 7 - 3(n-1)$ and find a_{20}.
2. An arithmetic sequence has $a_1 = 4$, $d = 5$. Calculate S_{30}.
3. A geometric sequence has $a_1 = 81$, $q = 1/3$. Find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Determine whether $5 + 5/2 + 5/4 + \cdots$ converges and find its sum.
7. A machine loses 12 percent of its value per year. Express its value after n years if it initially costs \$12,000.
8. For $a_{n+1} = (a_n + 10)/2$, $a_1 = 2$, find the fixed point and an explicit formula for a_n.

10 Synthesis: Motion and Patterns

Central question. How can the different mathematical languages of this course describe one evolving system?

10.1 One motion, several descriptions

Imagine a point traveling counterclockwise around a circle of radius R with angular speed ω. Its angle at time t is $\theta(t) = \theta_0 + \omega t$. In parametric form,

$$x(t) = R\cos(\theta_0 + \omega t), \qquad y(t) = R\sin(\theta_0 + \omega t).$$

Its polar form is $(r, \theta) = (R, \theta_0 + \omega t)$, and its complex form is

$$\boxed{z(t) = R(\cos(\theta_0 + \omega t) + i\sin(\theta_0 + \omega t))}.$$

These are not competing models: they encode the same physical position in different coordinate languages.

10.2 Velocity, vectors, and periodicity

Over a short time interval Δt, the displacement vector is

$$\Delta\mathbf{r} = \langle x(t + \Delta t) - x(t), y(t + \Delta t) - y(t)\rangle.$$

For constant angular speed the point completes a revolution after $T = 2\pi/|\omega|$ when $\omega \neq 0$. Its horizontal and vertical coordinates repeat with that period. The direction of motion is tangent to the circle and perpendicular to the radius: this is the geometric heart of circular motion.

10.3 Sampling creates a sequence

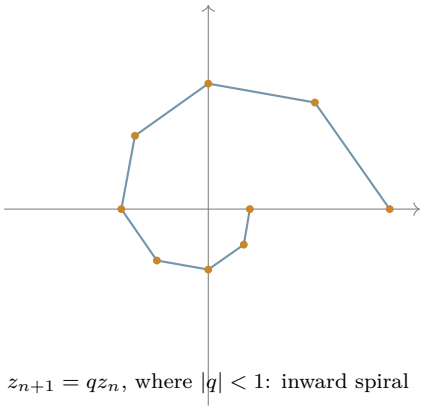
Suppose we record the position every Δt seconds. The resulting sequence of complex positions is

$$z_n = R(\cos(\theta_0 + n\delta) + i\sin(\theta_0 + n\delta)), \qquad \delta = \omega\Delta t.$$

Writing $u = \cos\delta + i\sin\delta$ gives

$$\boxed{z_{n+1} = uz_n, \qquad z_n = u^n z_0.}$$

De Moivre's theorem then explains every sample as a repeated rotation. If $\delta = 2\pi/m$ for a positive integer m, the sampled orbit closes after m steps (provided this is the smallest such positive step count). If $\delta/(2\pi)$ is irrational, no two different samples have exactly the same position.



$z_{n+1} = qz_n$, where $|q| < 1$: inward spiral

10.4 Scaling and rotation together

Now consider the recurrence $z_{n+1} = qz_n$ where

$$q = \rho(\cos\delta + i\sin\delta).$$

Its explicit solution is $z_n = q^n z_0$. Consequently,

$$|z_n| = \rho^n |z_0|, \qquad \arg z_n \equiv \arg z_0 + n\delta \pmod{2\pi}$$

for nonzero terms. If $0 < \rho < 1$, the points spiral toward the origin; if $\rho > 1$, they spiral outward; and if $\rho = 1$, they remain on a circle. Thus a geometric sequence and a rotation combine to produce a spiral.

10.5 Sums represent accumulated effects

If a system moves by step vectors $\mathbf{v}_0, \mathbf{v}_1, \ldots$, its total displacement after N steps is the vector sum $\sum_{n=0}^{N-1} \mathbf{v}_n$. For a complex-valued geometric pattern $\Delta z_n = aq^n$ with $q \neq 1$,

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q}.$$

Notice that the location sequence z_n and the step sequence Δz_n are different objects. Careful modeling starts by distinguishing what is being measured.

10.6 Different curves arise from different constraints

Circular and elliptical motion can be expressed parametrically. A projectile follows a parabola when horizontal velocity is constant and vertical acceleration is constant. Polar formulas express rays and radial patterns directly. The useful representation depends on whether we need a shape, a position, a direction, or a rule for change.

10.7 A complete example

A sensor begins at $z_0 = 2$ in the complex plane. Every second its controller multiplies the position by $q = \frac{1}{2}(\cos 60^\circ + i\sin 60^\circ)$. After n steps,

$$z_n = 2\left(\frac{1}{2}\right)^n (\cos(60^\circ n) + i\sin(60^\circ n)).$$

After three steps the modulus is $1/4$, and the direction is 180°. Hence $z_3 = -1/4$. The modulus obeys a geometric sequence, the direction obeys an arithmetic progression of angles, and the position follows a discrete inward spiral.

10.8 Exercises

1. Write Cartesian, polar, and complex formulas for a circle of radius 5 rotating once every 20 seconds, starting on the positive horizontal axis.

2. Sample that motion every five seconds. Give its first five complex positions and identify its repetition period in samples.

3. For $z_{n+1} = iz_n$ and $z_0 = 1 + i$, calculate $z_1, \ldots, z_4$ and describe the geometry.

4. For $z_{n+1} = (1/2)iz_n$, $z_0 = 4$, find z_4 and explain the change in magnitude and argument.

5. A point travels along $x = 3\cos t$, $y = 2\sin t$. Eliminate the parameter and describe one complete traversal.

6. Explain why a constant-speed circular trajectory is periodic but the spiral $z_n = (0.8e^{i\pi/3})^n z_0$ is not periodic when $z_0 \neq 0$.

7. Compute the finite sum $1 + i + i^2 + i^3$ and interpret its geometric cancellation.

8. Describe a real-world process that could be analyzed with at least three of the course's mathematical representations.